Customer Wait-Time Improvement Project

Project Type: Independent Systems Engineering Project

Problem: Customers at a small business may experience long wait times before receiving service.

Goal: Identify possible causes of waiting and develop a process improvement that could reduce wait times without adding employees.

Method: Map the current process, analyze simulated data using Google Sheets, identify bottlenecks, and compare a proposed improvement.

Tools: Google Docs, Google Sheets, diagrams.net.

Data: Simulated data created for educational purposes.

Possible Causes of Waiting:

1. Peak demand: too many customers arrive at the same time.
2. Long service time: some customers have longer transactions, which slows down the line for everyone behind them.

Proposed Improvement: Express Line

Quick transactions stay in the main teller line. Long transactions are sent to an existing staff member (such as a personal banker), so they don't block quick customers during busy times.

Results:

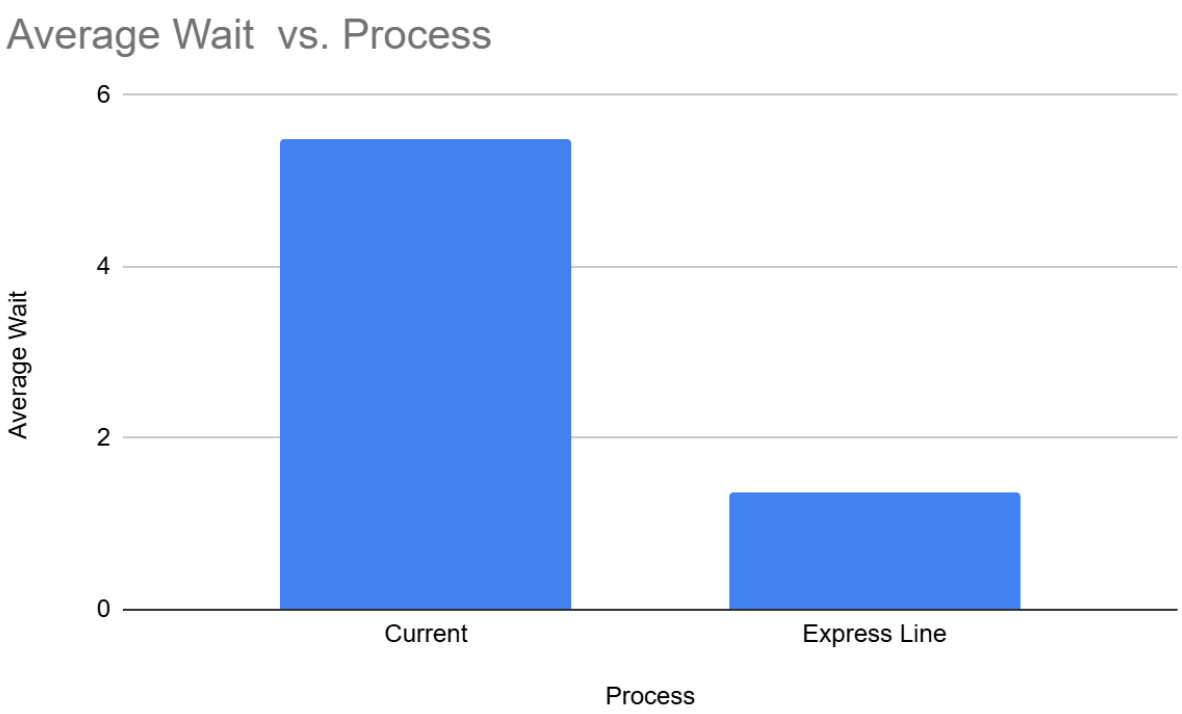
Current process average wait time: 5.5 minutes

Express line average wait time: 1.375 minutes

The express line reduced average wait time by 75% without adding employees.

Conclusion:

The main cause of long waits was long transactions blocking the line during busy times. An express line that sends long transactions to existing staff reduced average wait time from 5.5 minutes to 1.375 minutes. This shows that improving the process can reduce wait times without adding employees.


Average Wait vs. Process
6
4
2
0
Average Wait
Current
Express Line
Process